National
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# 2016 Mathematics

# Higher Paper 2

# Finalised Marking Instructions

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**General Marking Principles for Higher Mathematics**

*This information is provided to help you understand the general principles you must apply when marking candidate responses to questions in this Paper. These principles must be read in conjunction with the detailed marking instructions, which identify the key features required in candidate responses.*

*For each question the marking instructions are generally in two sections, namely Illustrative Scheme and Generic Scheme. The Illustrative Scheme covers methods which are commonly seen throughout the marking. The Generic Scheme indicates the rationale for which each mark is awarded. In general, markers should use the Illustrative Scheme and only use the Generic Scheme where a candidate has used a method not covered in the Illustrative Scheme.*

**(a)** Marks for each candidate response must always be assigned in line with these General Marking Principles and the Detailed Marking Instructions for this assessment.

**(b)** Marking should always be positive. This means that, for each candidate response, marks are accumulated for the demonstration of relevant skills, knowledge and understanding: they are not deducted from a maximum on the basis of errors or omissions.

**(c)** If a specific candidate response does not seem to be covered by either the principles or detailed Marking Instructions, and you are uncertain how to assess it, you must seek guidance from your Team Leader.

**(d)** Credit must be assigned in accordance with the specific assessment guidelines.

**(e)** One mark is available for each •. There are no half marks.

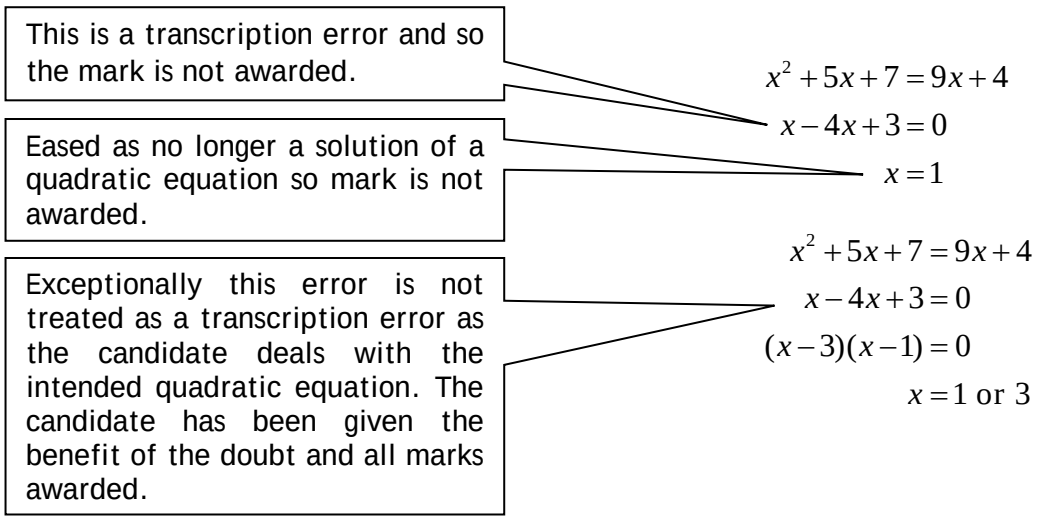
**(f)** Working subsequent to an error must be **followed through**, with possible credit for the subsequent working, provided that the level of difficulty involved is approximately similar. Where, subsequent to an error, the working for a follow through mark has been eased, the follow through mark cannot be awarded.

**(g)** As indicated on the front of the question paper, full credit should only be given where the solution contains appropriate working. Unless specifically mentioned in the marking instructions, a correct answer with no working receives no credit.

**(h)** Candidates may use any mathematically correct method to answer questions except in cases where a particular method is specified or excluded.

**(i)** As a consequence of an error perceived to be trivial, casual or insignificant, eg $6\times 6=12$ candidates lose the opportunity of gaining a mark. However, note the second example in comment **(j)**.

**(j)** Where a transcription error (paper to script or within script) occurs, the candidate should normally lose the opportunity to be awarded the next process mark, eg




**(k) Horizontal/vertical marking**

Where a question results in two pairs of solutions, this technique should be applied, but only if indicated in the detailed marking instructions for the question.

Example:

| | $\bullet^5$ | $\bullet^6$ |
|---|---|---|
| $\bullet^5$ | $x=2$ | $x=-4$ |
| $\bullet^6$ | $y=5$ | $y=-7$ |

Horizontal: $\bullet^5$ $x=2$ and $x=-4$ Vertical: $\bullet^5$ $x=2$ and $y=5$
$\bullet^6$ $y=5$ and $y=-7$ $\bullet^6$ $x=-4$ and $y=-7$

Markers should choose whichever method benefits the candidate, but **not** a combination of both.

**(l)** In final answers, unless specifically mentioned in the detailed marking instructions, numerical values should be simplified as far as possible, eg:

$\frac{15}{12}$ must be simplified to $\frac{5}{4}$ or $1\frac{1}{4}$ $\frac{43}{1}$ must be simplified to $43$

$\frac{15}{0\cdot3}$ must be simplified to $50$ $\frac{4/5}{3}$ must be simplified to $\frac{4}{15}$

$\sqrt{64}$ must be simplified to $8$*

*The square root of perfect squares up to and including $100$ must be known.

**(m)** Commonly Observed Responses (COR) are shown in the marking instructions to help mark common and/or non-routine solutions. CORs may also be used as a guide when marking similar non-routine candidate responses.

**(n)** Unless specifically mentioned in the marking instructions, the following should not be penalised:

- Working subsequent to a correct answer
- Correct working in the wrong part of a question
- Legitimate variations in numerical answers/algebraic expressions, eg angles in degrees rounded to nearest degree
- Omission of units
- Bad form (bad form only becomes bad form if subsequent working is correct), eg $(x^3+2x^2+3x+2)(2x+1)$ written as $(x^3+2x^2+3x+2)\times 2x+1$

  $2x^4+4x^3+6x^2+4x+x^3+2x^2+3x+2$ written as $2x^4+5x^3+8x^2+7x+2$ gains full credit

- Repeated error within a question, but not between questions or papers

**(o)** In any 'Show that…' question, where the candidate has to arrive at a required result, the last mark of that part is not available as a follow-through from a previous error unless specified in the detailed marking instructions.

**(p)** All working should be carefully checked, even where a fundamental misunderstanding is apparent early in the candidate's response. Marks may still be available later in the question so reference must be made continually to the marking instructions. The appearance of the correct answer does not necessarily indicate that the candidate has gained all the available marks.

**(q)** Scored-out working which has not been replaced should be marked where still legible. However, if the scored out working has been replaced, only the work which has not been scored out should be marked.

**(r)** Where a candidate has made multiple attempts using the same strategy and not identified their final answer, mark all attempts and award the lowest mark.

For example:

| | |
|---|---|
| Strategy 1 attempt 1 is worth 3 marks. | Strategy 2 attempt 1 is worth 1 mark. |
| Strategy 1 attempt 2 is worth 4 marks. | Strategy 2 attempt 2 is worth 5 marks. |
| From the attempts using strategy 1, the resultant mark would be 3. | From the attempts using strategy 2, the resultant mark would be 1. |

In this case, award 3 marks.

**Specific Marking Instructions for each question**

| Question | | | Generic Scheme | Illustrative Scheme | Max Mark |
|---|---|---|---|---|---|
| **1.** | (a) | i | $\bullet^1$ state the midpoint $\mathrm{M}$ | $\bullet^1$ $(2,4)$ | **1** |
| | | **ii** | $\bullet^2$ calculate gradient of median<br><br>$\bullet^3$ determine equation of median | $\bullet^2$ $4$<br><br>$\bullet^3$ $y=4x-4$ | **2** |

**Notes:**

1. $\bullet^3$ is not available as a consequence of using a perpendicular gradient.
2. Accept any rearrangement of $y=4x-4$ for $\bullet^3$.
3. On this occasion, accept $y-4=4(x-2)$ or $y-(-4)=4(x-0)$; however, in future candidates should expect that the final equation will only be accepted when it involves a single constant term.
4. $\bullet^3$ is only available as a consequence of using points M and P, or any other point which lies on PM, for example the midpoint $(1,0)$ .

**Commonly Observed Responses:**

| | | | | | |
|---|---|---|---|---|---|
| | (b) | | $\bullet^1$ calculate gradient of PR<br><br>$\bullet^2$ use property of perpendicular lines<br><br>$\bullet^3$ determine equation of line | $\bullet^1$ $1$<br><br>$\bullet^2$ $-1$ stated or implied by $\bullet^6$<br><br>$\bullet^3$ $y=-x+6$ | **3** |

**Notes:**

5. $\bullet^6$ is only available as a consequence of using $\mathrm{M}$ and a perpendicular gradient.
6. Candidates who use a gradient perpendicular to QR cannot gain $\bullet^4$ but $\bullet^5$ and $\bullet^6$ are still available. See Candidate A.
7. Beware of candidates who use the coordinates of P and Q to arrive at $m=-1$. See Candidate B.
8. On this occasion, accept $y-4=-1(x-2)$; however, in future candidates should expect that the final equation will only be accepted when it involves a single constant term.

| Question | Generic Scheme | Illustrative Scheme | Max Mark |
|---|---|---|---|
| **Commonly Observed Responses:** | | | |
| **Candidate A**<br>$m_{QR}=\frac{1}{4}$ $\bullet^4$ ✗ $m_{perp}=-4$ $\bullet^5$ ✓1<br>$y=-4x+12$ $\bullet^6$ ✓1 | | **Candidate B – BEWARE**<br>$m_{PQ}=\frac{2-(-4)}{-6-0}$ $\bullet^4$ ^<br>$=-1$ $\bullet^5$ ✗<br>$y-4=-1(x-2)$ $\bullet^6$ ✓2<br>$y=-x+6$<br>Note: $\bullet^4$ $\bullet^5$ and $\bullet^6$ may still be available for any candidate that demonstrates that PQ is also perpendicular to PR. | |
| (c) | $\bullet^1$ find the midpoint of PR<br>$\bullet^2$ substitute $x$-coordinate into equation of L.<br>$\bullet^3$ verify $y$-coordinate and communicate conclusion | **Method 1**<br>$\bullet^1$ $(5,1)$<br>$\bullet^2$ $y=-5+6$ $(1=-x+6)$<br>$\bullet^3$ $y=1(x=5)$ $\therefore$ L passes through the midpoint of PR | **3** |
| | $\bullet^7$ find the midpoint of PR<br>$\bullet^8$ substitute $x$ and $y$ coordinates into the equation of L<br>$\bullet^9$ verify the point satisfies the equation and communicate conclusion | **Method 2**<br>$\bullet^7$ $x+y=6$ sub $(5,1)$<br>$\bullet^8$ $5+1=6$<br>$\bullet^9$ $\therefore$ point $(5,1)$ satisfies equation. | |
| | $\bullet^7$ find the midpoint of PR<br>$\bullet^8$ find equation of PR<br>$\bullet^9$ use simultaneous equations and communicate conclusion | **Method 3**<br>$\bullet^7$ $(5,1)$<br>$\bullet^8$ $y=x-4$<br>$\bullet^9$ $y=1,\ x=5$ $\therefore$ L passes through the midpoint of PR | |

| Question | Generic Scheme | Illustrative Scheme | Max Mark |
|---|---|---|---|
| | | **Method 4** | |
| | •7 find the midpoint of PR | •7 $(5,1)$ | |
| | •8 find equation of perpendicular bisector of PR | •8 $y-1=-1(x-5)\rightarrow y=-x+6$ | |
| | •9 communicate conclusion | •9 The equation of the perpendicular bisector is the same as L therefore L passes through the midpoint of PR. | |

**Notes:**

9. A relevant statement is required for •9 to be awarded.
10. Erroneous working accompanied by a statement such as "L does not pass through the midpoint." does NOT gain •9.
11. Beware of candidates substituting $(1,5)$ instead of $(5,1)$
12. On this occasion, for Method 3, at •8 accept $y-1=1(x-5)$; however, in future candidates should expect that the final equation will only be accepted when it involves a single constant term.

**Commonly Observed Responses:**

**Candidate C**

$(5,1)$ mid-point •7 ✓

$y+x=6$

Sub $(5,1)$ •8 ×

$5+1=6$

$\therefore$ point $(5,1)$ satisfies equation. •9 ×

Candidate has substituted 5 for $y$ and 1 for $x$.

| Question | Generic Scheme | Illustrative Scheme | Max Mark |
|---|---|---|---|
| **2.** | $\bullet^1$ use the discriminant<br><br>$\bullet^2$ simplify and apply the condition for no real roots<br><br>$\bullet^3$ state range | $\bullet^1$ $(-2)^2-4(1)(3-p)$<br><br>$\bullet^2$ $-8+4p<0$<br><br>$\bullet^3$ $p<2$ | **3** |

**Notes:**

1. At the $\bullet^1$ stage, treat $(-2)^2-4(1)3-p$ and $-2^2-4(1)(3-p)$ as bad form only if the candidate deals with the 'bad form' **term** correctly in the inequation at $\bullet^2$.
2. If candidates have the condition 'discriminant = 0', then $\bullet^2$ and $\bullet^3$ are unavailable.
3. If candidates have the condition 'discriminant $>0$', 'discriminant $\geq 0$' or 'discriminant $\leq 0$' then $\bullet^2$ is lost, but $\bullet^3$ is available **provided the discriminant has been simplified correctly at** $\bullet^2$.
4. If a candidate works with an equation, then $\bullet^2$ and $\bullet^3$ are unavailable. However, see Candidate D.

**Commonly Observed Responses:**

**Candidate A**

$(-2)^2-4(1)3-p$ $\bullet^1$ ✓
$-8+4p<0$ $\bullet^2$ ✓
$p<2$ $\bullet^3$ ✓

**Candidate B**

$(-2)^2-4(1)(3-p)$ $\bullet^1$ ✓
$-8-4p<0$ $\bullet^2$ ×
$p>-2$ $\bullet^3$ ✓1

**Candidate C**

$(-2)^2-4(1)3-p$ $\bullet^1$ ×
$-8-p<0$ $\bullet^2$ ✓2 eased
$p>-8$ $\bullet^3$ ✓2 eased

**Candidate D – Special Case**

$b^2-4ac<0$
$(-2)^2-4(1)(3-p)=0$ $\bullet^1$ ✓
$-8+4p=0$ $\bullet^2$ ✓
$p=2$
$p<2$ $\bullet^3$ ✓2

$\bullet$2 is awarded since the condition (first line), its application (final line) and the simplification of the discriminant all appear.

**Candidate E**

$-2^2-4(1)(3-p)$ $\bullet^1$ ✓
$-8+4p<0$ $\bullet^2$ ✓
$p<2$ $\bullet^3$ ✓

**Candidate F**

$-2^2-4(1)(3-p)$ $\bullet^1$ ×
$-16+4p<0$ $\bullet^2$ ✓2
$p<4$ $\bullet^3$ ✓1

**Candidate G**

$-2^2-4(1)(3-p)=0$ $\bullet^1$ ✓
$-8+4p=0$ $\bullet^2$ ×
$p=2$ $\bullet^3$ ×

| Question | | | Generic Scheme | Illustrative Scheme | Max Mark |
|---|---|---|---|---|---|
| **3.** | (a) | i | $\bullet^1$ know to substitute $x=-1$<br><br>$\bullet^2$ complete evaluation, interpret result and state conclusion | **Method 1**<br>$\bullet^1$ $2(-1)^3-9\times(-1)^2+3\times(-1)+14$<br><br>$\bullet^2$ $=0$ $\therefore$ $(x+1)$ is a factor | **2** |
| | | | $\bullet^1$ know to use $x=-1$ in synthetic division<br><br>$\bullet^2$ complete division, interpret result and state conclusion | **Method 2**<br>$\bullet^1$ $-1 \mid 2 \quad -9 \quad 3 \quad 14$<br>$\quad\quad\quad -2$<br>$\quad 2 \quad -11$<br><br>$\bullet^2$ $-1 \mid 2 \quad -9 \quad 3 \quad 14$<br>$\quad\quad\quad -2 \quad 11 \quad -14$<br>$\quad 2 \quad -11 \quad 14 \quad 0$<br>remainder = 0 $\therefore$ $(x+1)$ is a factor | |
| | | | $\bullet^1$ start long division and find leading term in quotient<br><br>$\bullet^2$ complete division, interpret result and state conclusion | **Method 3**<br>$\bullet^1$ $(x+1) \overline{)\, 2x^3-9x^2+3x+14}$ with $2x^2$ above<br><br>$\bullet^2$ $(x+1) \overline{)\, 2x^3-9x^2+3x+14}$ with $2x^2-11x+14$ above<br>$2x^3+2x^2$<br>$-11x^2+3x$<br>$-11x^2-11x$<br>$14x+14$<br>$14x+14$<br>$0$<br>remainder = 0 $\therefore$ $(x+1)$ is a factor | |

| Question | Generic Scheme | Illustrative Scheme | Max Mark |
|---|---|---|---|

**Notes:**

1. Communication at $\bullet^2$ must be consistent with working at that stage ie a candidate's working must arrive legitimately at 0 before $\bullet^2$ can be awarded.
2. Accept any of the following for $\bullet^2$ :
   - '$f(-1)=0$ so $(x+1)$ is a factor'
   - 'since remainder = $0$, it is a factor'
   - the 0 from any method linked to the word 'factor' by eg 'so', 'hence', '$\therefore$', '$\rightarrow$', '$\Rightarrow$'
3. Do not accept any of the following for $\bullet^2$ :
   - double underlining the zero or boxing the zero without comment
   - '$x=1$ is a factor', '$(x-1)$ is a factor', '$(x-1)$ is a root', '$x=-1$ is a root', '$(x+1)$ is a root'
   - the word 'factor' only with no link

**Commonly Observed Responses:**

| | | | | | |
|---|---|---|---|---|---|
| | | ii | $\bullet^3$ state quadratic factor<br><br>$\bullet^4$ find remaining linear factors or substitute into quadratic formula<br><br>$\bullet^5$ state solution | $\bullet^3$ $2x^2-11x+14$<br><br>$\bullet^4$ $....(2x-7)(x-2)$<br>or<br>$\frac{11\pm\sqrt{(-11)^2-4\times2\times14}}{2\times2}$<br><br>$\bullet^5$ $x=-1,\ 2,\ 3{\cdot}5$ | **3** |

**Notes:**

4. On this occasion, the appearance of "$=0$" is not required for $\bullet^5$ to be awarded.
5. Be aware that the solution, $x=-1,\ 2,\ 3{\cdot}5$, may not appear until part (b).

**Commonly Observed Responses:**

| Question | | | Generic Scheme | Illustrative Scheme | Max Mark |
|---|---|---|---|---|---|
| | (b) | (i) | $\bullet^6$ state coordinates | $\bullet^6$ $(-1,0)$ and $(2,0)$ | **1** |
| **Notes:** | | | | | |
| 6. '$-1$ and $2$' does not gain $\bullet^6$<br>7. $x=-1, y=0$ **and** $x=2, y=0$ gains $\bullet^6$ | | | | | |
| **Commonly Observed Responses:** | | | | | |
| | | | | | |
| | | (ii) | $\bullet^7$ know to integrate with respect to $x$<br>$\bullet^8$ integrate<br>$\bullet^9$ interpret limits and substitute<br>$\bullet^{10}$ evaluate integral | $\bullet^7$ $\int\left(2x^3-9x^2+3x+14\right)dx$<br>$\bullet^8$ $\frac{2x^4}{4}-\frac{9x^3}{3}+\frac{3x^2}{2}+14x$<br>$\bullet^9$ $\left(\frac{2\times2^4}{4}-\frac{9\times2^3}{3}+\frac{3\times2^2}{2}+14\times2\right)-\left(\frac{2\times(-1)^4}{4}-\frac{9\times(-1)^3}{3}+\frac{3\times(-1)^2}{2}+14\times(-1)\right)$<br>$\bullet^{10}$ 27 | **4** |
| | | | **Candidate A**<br>$\int\left(2x^3-9x^2+3x+14\right)dx$<br>$\frac{2x^4}{4}-\frac{9x^3}{3}+\frac{3x^2}{2}+14x$<br>27 | $\bullet^7$ ✓<br>$\bullet^8$ ✓<br>$\bullet^9$ ^ $\bullet^{10}$ ✓1 | |
| | | | **Candidate B**<br>$\int\left(2x^3-9x^2+3x+14\right)dx$<br>$\frac{2x^4}{4}-\frac{9x^3}{3}+\frac{3x^2}{2}+14x$<br>$\left(\frac{2\times(-1)^4}{4}-\frac{9\times(-1)^3}{3}+\frac{3\times(-1)^2}{2}+14\times(-1)\right)-\left(\frac{2\times2^4}{4}-\frac{9\times2^3}{3}+\frac{3\times2^2}{2}+14\times2\right)$<br>$-27$, hence area is 27<br>**However**, $-27=27$ does not gain $\bullet^{10}$. | $\bullet^7$ ✓<br>$\bullet^8$ ✓<br>$\bullet^9$ ×<br>$\bullet^{10}$ ✓1 | |

| Question | Generic Scheme | Illustrative Scheme | Max Mark |
|---|---|---|---|
| | **Candidate C**<br>$\int ... = -27$ cannot be negative so $= 27$ $\bullet^{10}$ ✗<br>However, $\int ... = -27$ so Area $= 27$ $\bullet^{10}$ ✓ | | |
| **Notes:** | | | |
| 8. $\bullet^7$ is not available to candidates who omit '$dx$'<br>9. Do not penalise the absence of brackets at the $\bullet^7$ stage<br>10. Where a candidate differentiates one or more terms at $\bullet^8$, then $\bullet^8$, $\bullet^9$ and $\bullet^{10}$ are not available.<br>11. Candidates who substitute limits without integrating do not gain $\bullet^8$, $\bullet^9$ or $\bullet^{10}$.<br>12. For candidates who make an error in (a), $\bullet^9$ is available only if the lower limit is negative and the upper limit is positive.<br>13. Do not penalise the inclusion of '+c'. | | | |
| **Commonly Observed Responses:** | | | |
| | | | |

| Question | | | Generic Scheme | Illustrative Scheme | Max Mark |
|---|---|---|---|---|---|
| **4.** | (a) | | $\bullet^1$ centre of $C_1$<br>$\bullet^2$ radius of $C_1$<br>$\bullet^3$ centre of $C_2$<br>$\bullet^4$ radius of $C_2$ | $\bullet^1$ $(-5,6)$<br>$\bullet^2$ $3$<br>$\bullet^3$ $(3,0)$<br>$\bullet^4$ $5$ | **4** |
| **Notes:** | | | | | |
| | | | | | |
| **Commonly Observed Responses:** | | | | | |
| | | | | | |
| | (b) | | $\bullet^5$ calculate the distance between the centres<br>$\bullet^6$ calculate the sum of the radii<br>$\bullet^7$ interpret significance of calculations | $\bullet^1$ $10$<br>$\bullet^2$ $8$<br>$\bullet^3$ $8<10 \therefore$ the circles do not intersect | **3** |
| **Notes:** | | | | | |
| 1. For $\bullet^7$ to be awarded a comparison must appear.<br>2. Candidates who write '$r_1+r_2<\mathrm{D}$', or similar, must have identified the value of $r_1+r_2$ and the value of $\mathrm{D}$ somewhere in their solution for $\bullet^7$ to be awarded.<br>3. Where earlier errors lead to the candidate dealing with non-integer values, do not penalise inaccuracies in rounding unless they lead to an inconsistent conclusion. | | | | | |
| **Commonly Observed Responses:** | | | | | |
| | | | | | |

| Question | | | Generic Scheme | Illustrative Scheme | Max Mark |
|---|---|---|---|---|---|
| **5.** | (a) | | $\bullet^1$ find $\overrightarrow{AB}$<br><br>$\bullet^2$ find $\overrightarrow{AC}$ | $\bullet^1$ $\begin{pmatrix}-8\\16\\2\end{pmatrix}$<br><br>$\bullet^2$ $\begin{pmatrix}-2\\-8\\16\end{pmatrix}$ | **2** |
| **Notes:** | | | | | |
| 1. For candidates who find **both** $\overrightarrow{BA}$ **and** $\overrightarrow{CA}$ correctly, only $\bullet^2$ is available (repeated error).<br>2. Accept vectors written horizontally. | | | | | |
| **Commonly Observed Responses:** | | | | | |
| | | | | | |
| | (b) | | **Method 1**<br><br>$\bullet^1$ evaluate $\overrightarrow{AB}.\overrightarrow{AC}$<br><br>$\bullet^2$ evaluate $\lvert\overrightarrow{AB}\rvert$ and $\lvert\overrightarrow{AC}\rvert$<br><br>$\bullet^3$ use scalar product<br><br>$\bullet^4$ calculate angle | **Method 1**<br><br>$\bullet^1$ $\overrightarrow{AB}.\overrightarrow{AC}=16-128+32=-80$<br><br>$\bullet^2$ $\lvert\overrightarrow{AB}\rvert=\lvert\overrightarrow{AC}\rvert=18$<br><br>$\bullet^3$ $\cos BAC=\dfrac{-80}{18\times18}$<br><br>$\bullet^4$ $104\cdot3°$ or $1\cdot82$ radians | **4** |
| | | | **Method 2**<br><br>$\bullet^3$ calculate length of $BC$<br><br>$\bullet^4$ calculate lengths of $AB$ and $AC$<br><br>$\bullet^5$ use cosine rule<br><br>$\bullet^6$ calculate angle | **Method 2**<br><br>$\bullet^3$ $BC=\sqrt{808}$<br><br>$\bullet^4$ $AB=AC=18$<br><br>$\bullet^5$ $\cos BAC=\dfrac{18^2+18^2-\sqrt{808}^2}{2\times18\times18}$<br><br>$\bullet^6$ $104\cdot3°$ or $1\cdot82$ radians | |

| Question | Generic Scheme | Illustrative Scheme | Max Mark |
|---|---|---|---|

**Notes:**

3. Accept $\sqrt{324}$ at $\bullet^4$ and $\bullet^5$.
4. $\bullet^5$ is not available to candidates who simply state the formula $\cos\theta = \frac{a.b}{|a||b|}$.

   However $\cos\theta = \frac{-80}{18\times 18}$ is acceptable. Similarly for Method 2.
5. Accept correct answers rounded to $104°$ or $1{\cdot}8$ radians.
6. Due to $\overrightarrow{AB}$ and $\overrightarrow{AC}$ having equal magnitude, $\bullet^4$ is not available unless both $|\overrightarrow{AB}|$ **and** $|\overrightarrow{AC}|$ have been stated.
7. $\bullet^6$ is only available as a result of using a valid strategy.
8. $\bullet^6$ is only available for a single angle.
9. For a correct answer with no working award 0/4.

**Commonly Observed Responses:**

**Candidate A**

$\overrightarrow{BA}.\overrightarrow{AC} = -16+128-32 = 80$ $\bullet^3$ ✗

$|\overrightarrow{AB}| = |\overrightarrow{AC}| = 18$ $\bullet^4$ ✓1

$\cos\theta = \frac{80}{18\times 18}$ $\bullet^5$ ✓1

$\bullet^6$ ✗

$75{\cdot}7$ or $1{\cdot}32$ radians

| Question | | | Generic Scheme | Illustrative Scheme | Max Mark |
|---|---|---|---|---|---|
| **6.** | (a) | | $\bullet^1$ state the number | $\bullet^1$ $200$ | **1** |
| **Notes:** | | | | | |
| | | | | | |
| **Commonly Observed Responses:** | | | | | |
| | | | | | |
| | (b) | | $\bullet^2$ interpret context and form equation<br><br>$\bullet^3$ knowing to use logarithms appropriately.<br><br>$\bullet^4$ simplify<br><br>$\bullet^5$ evaluate $t$ | $\bullet^2$ $2 = e^{0{\cdot}107t}$<br><br>$\bullet^3$ $\ln 2 = \ln\left(e^{0{\cdot}107t}\right)$<br><br>$\bullet^4$ $\ln 2 = 0{\cdot}107t$<br><br>$\bullet^5$ $t = 6{\cdot}478\ldots$ | **4** |

**Notes:**

1. Accept $400 = 200e^{0{\cdot}107t}$ or equivalent for $\bullet^2$
2. Any base may be used at the $\bullet^3$ stage.
3. $\bullet^3$ may be assumed by $\bullet^4$.
4. Accept $t = 6{\cdot}5$.
5. At $\bullet^5$ ignore incorrect units. However, see Candidates B and C.
6. The calculation at $\bullet^5$ must involve the evaluation of a logarithm within a valid strategy for $\bullet^5$ to be awarded.
7. Candidates who take an iterative approach to arrive at $t = 6{\cdot}5$ gain $\bullet^2$ only.
   However, if, in the iterations, $B(t)$ is evaluated for $t = 6{\cdot}45$ and $t = 6{\cdot}55$ then award 4/4.

**Commonly Observed Responses:**

**Candidate A**

$2 = e^{0{\cdot}107t}$ $\bullet^2$ ✓

$\log_{10} 2 = \log_{10}\left(e^{0{\cdot}107t}\right)$ $\bullet^3$ ✓

$\log_{10} 2 = 0{\cdot}107t \log_{10} e$ $\bullet^4$ ✓

$t = 6{\cdot}478\ldots$ $\bullet^5$ ✓

**Candidate B**

$t = 6{\cdot}48$ hours $\bullet^5$ ✓

$t = 6$ hours 48 minutes

**Candidate C**

$\ln(2) = 0{\cdot}107t$ $\bullet^4$ ✓

$t = 6$ hours 48 minutes $\bullet^5$ ×

**Candidate D**

$400 = 200e^{0{\cdot}107t}$ $\bullet^2$ ✓

$e^{0{\cdot}107t} = 2$ $\bullet^3$ ^

$t = 6{\cdot}48$ hours ✓1 $\bullet^4$ ✓1 $\bullet^5$

| Question | | | Generic Scheme | Illustrative Scheme | Max Mark |
|---|---|---|---|---|---|
| **7.** | (a) | | $\bullet^1$ expression for length in terms of $x$ and $y$<br>$\bullet^2$ obtain an expression for $y$<br>$\bullet^3$ demonstrate result | $\bullet^1$ $9x+8y$<br>$\bullet^2$ $y=\frac{108}{6x}$<br>$\bullet^3$ $\mathrm{L}(x)=9x+8\left(\frac{108}{6x}\right)$<br>leading to $\mathrm{L}(x)=9x+\frac{144}{x}$ | **3** |
| **Notes:** | | | | | |
| 1. The substitution for $y$ at $\bullet^3$ must be clearly shown for $\bullet^3$ to be available.<br>2. For candidates who omit some, or all, of the internal fencing, only $\bullet^2$ is available. | | | | | |
| **Commonly Observed Responses:** | | | | | |
| | | | | | |
| | (b) | | $\bullet^4$ know to and start to differentiate<br>$\bullet^5$ complete differentiation<br>$\bullet^6$ set derivative equal to 0<br>$\bullet^7$ obtain for $x$<br>$\bullet^8$ verify nature of stationary point<br>$\bullet^9$ interpret and communicate result | $\bullet^4$ $\mathrm{L}'(x)=9\ldots$<br>$\bullet^5$ $L'(x)=9-\frac{144}{x^2}$<br>$\bullet^6$ $9-\frac{144}{x^2}=0$<br>$\bullet^7$ $x=4$<br>$\bullet^8$ Table of signs for a derivative – see the additional page.<br>$\bullet^9$ Minimum at $x=4$<br>**or**<br>$\bullet^8$ $\mathrm{L}''(x)=\frac{288}{x^3}$<br>$\bullet^9$ $\mathrm{L}''(4)>0$ $\therefore$ minimum<br>Do not accept $\frac{d^2y}{dx^2}=\ldots$ | **6** |

| Question | Generic Scheme | Illustrative Scheme | Max Mark |
|---|---|---|---|

**Notes:**

3. For candidates who integrate at the $\bullet^4$ stage $\bullet^5$, $\bullet^6$, $\bullet^7$, $\bullet^8$ and $\bullet^9$ are unavailable.
4. $\bullet^7$, $\bullet^8$ and $\bullet^9$ are only available for working with a derivative which contains a term with an index $\leq -2$.
5. At $\bullet^5$ and $\bullet^6$ accept $-144x^{-2}$ .
6. $\sqrt{\frac{144}{9}}$ must be simplified at the $\bullet^7$, $\bullet^8$ or $\bullet^9$ stage for $\bullet^7$ to be awarded.
7. $\bullet^9$ is not available to candidates who consider a value $\leq 0$ in the neighbourhood of 4.
8. A candidate's table of signs must be valid and legitimately lead to a minimum for $\bullet^9$ to be awarded.
9. $\bullet^9$ is not available to candidates who state the minimum exists at a negative value of $x$.

**Commonly Observed Responses:**

**Table of signs for a derivative**

**Accept**

| $x$ | $-4^-$ | $-4$ | $-4^+$ |
|---|---|---|---|
| $L'(x)$ | + | 0 | − |
| Shape or Slope |  | | |

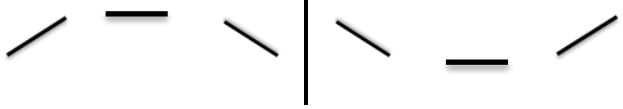
| $x$ | $4^-$ | $4$ | $4^+$ |
|---|---|---|---|
| $L'(x)$ | − | 0 | + |
| Shape or Slope |  | | |

| $x$ | $\rightarrow$ | $-4$ | $\rightarrow$ |
|---|---|---|---|
| $L'(x)$ | + | 0 | − |
| Shape or Slope |  | | |

| $x$ | $\rightarrow$ | $4$ | $\rightarrow$ |
|---|---|---|---|
| $L'(x)$ | − | 0 | + |
| Shape or Slope |  | | |

Here, for exemplification, tables of signs considering both roots separately have been displayed. However, in this question, it was only expected that candidates would consider the positive root for $\bullet^8$. Do not penalise the consideration of the negative root.

Arrows are taken to mean "in the neighbourhood of"

| $x$ | $a$ | $-4$ | $b$ | $c$ | $4$ | $d$ |
|---|---|---|---|---|---|---|
| $L'(x)$ | + | 0 | − | − | 0 | + |
| Shape or Slope |  | | | | | |

Where: $a < -4$ , $-4 < b < 0$ , $0 < c < 4$ , $d > 4$

**Do not Accept**

| $x$ | $a$ | $-4$ | $b$ | $4$ | $c$ |
|---|---|---|---|---|---|
| $L'(x)$ | + | 0 | − | 0 | + |
| Shape or Slope |  | | | | |

Since the function is discontinuous '$-4 < b < 4$' is not acceptable.

| $x$ | $\rightarrow$ | $-4$ | $\rightarrow$ | $4$ | $\rightarrow$ |
|---|---|---|---|---|---|
| $L'(x)$ | + | 0 | − | 0 | + |
| Shape or Slope |  | | | | |

Since the function is discontinuous '$-4 \rightarrow 4$' is not acceptable.

**General Comments**

- For this question do not penalise the omission of '$x$' or the word 'shape'/'slope'.
- Stating values of $L'(x)$ in the table is an acceptable alternative to writing '+' or '-' signs.
- Acceptable alternatives for $L'(x)$ are: $L'$ , $\frac{dL}{dx}$ or $9-\frac{144}{x^2}$ . **DO NOT** accept $\frac{dy}{dx}$ or $f'(x)$.

| Question | | | Generic Scheme | Illustrative Scheme | Max Mark |
|---|---|---|---|---|---|
| **8.** | (a) | | $\bullet^1$ use compound angle formula<br><br>$\bullet^2$ compare coefficients<br><br>$\bullet^3$ process for $k$<br><br>$\bullet^4$ process for $a$ and express in required form | $\bullet^1$ $k\cos x\cos a - k\sin x\sin a$ **stated explicitly**<br>$\bullet^2$ $k\cos a = 5, k\sin a = 2$ **stated explicitly**<br>$\bullet^3$ $k = \sqrt{29}$<br>$\bullet^4$ $\sqrt{29}\cos(x + 0\cdot38)$ | **4** |

**Notes:**

1. Treat $k\cos x\cos a - \sin x\sin a$ as bad form only if the equations at the $\bullet^2$ stage both contain $k$.
2. $\sqrt{29}\cos x\cos a - \sqrt{29}\sin x\sin a$ or $\sqrt{29}(\cos x\cos a - \sin x\sin a)$ is acceptable for $\bullet^1$ and $\bullet^3$.
3. Accept $k\cos a = 5, -k\sin a = -2$ for $\bullet^2$.
4. $\bullet^2$ is not available for $k\cos x = 5$, $k\sin x = 2$, however, $\bullet^4$ is still available.
5. $\bullet^3$ is only available for a single value of $k$, $k > 0$.
6. Candidates who work in degrees and do not convert to radian measure do not gain $\bullet^4$.
7. Candidates may use any form of the wave equation for $\bullet^1$, $\bullet^2$ and $\bullet^3$, however, $\bullet^4$ is only available if the value of $a$ is interpreted in the form $k\cos(x + a)$.
8. Accept any answer for $a$ that rounds to $0\cdot38$.
9. Evidence for $\bullet^4$ may not appear until part (b).

**Commonly Observed Responses:**

**Responses with missing information in working:**

**Candidate A**

$\bullet^1$ ^

$\sqrt{29}\cos a = 5$ $\bullet^2$ ✓

$\sqrt{29}\sin a = 2$ $\bullet^3$ ✓

$\tan a = \frac{2}{5}$, $a = 0\cdot3805...$

$\sqrt{29}\cos(x + 0\cdot38)$ $\bullet^4$ ✓

**Candidate B**

$k\cos x\cos a - k\sin x\sin a$ $\bullet^1$ ✓

$\cos a = 5$

$\sin a = 2$ $\bullet^2$ ×

$\bullet^3$ ✓

$\tan a = \frac{2}{5}$

$a = 0\cdot38$ — Not consistent with equations at $\bullet^2$.

$\sqrt{29}\cos(x + 0\cdot38)$ $\bullet^4$ ×

**Responses with the correct expansion of $k\cos(x + a)$ but errors for either $\bullet^2$ or $\bullet^4$.**

**Candidate C**

$k\cos a = 5, k\sin a = 2$ $\bullet^2$ ✓

$\tan a = \frac{5}{2}$ $\bullet^4$ ×

$a = 1\cdot19$

**Candidate D**

$k\cos a = 2, k\sin a = 5$ $\bullet^2$ ×

$\tan a = \frac{5}{2}$, $a = 1\cdot19$

$\sqrt{29}\cos(x + 1\cdot19)$ $\bullet^4$ ✓1

**Candidate E**

$k\cos a = 5, k\sin a = -2$ $\bullet^2$ ×

$\tan a = \frac{-2}{5}$

$\sqrt{29}\cos(x + 5\cdot90)$ $\bullet^4$ ✓1

| Question | Generic Scheme | Illustrative Scheme | Max Mark |
|---|---|---|---|

**Responses with the incorrect labelling;** $k\cos A\cos B - k\sin A\sin B$ from formula list.

| Candidate F | Candidate G | Candidate H |
|---|---|---|
| $k\cos A\cos B - k\sin A\sin B$ $\bullet^1$✘ | $k\cos A\cos B - k\sin A\sin B$ $\bullet^1$✘ | $k\cos A\cos B - k\sin A\sin B$ $\bullet^1$✘ |
| $k\cos a = 5,\ k\sin a = 2$ $\bullet^2$✓ | $k\cos x = 5,\ k\sin x = 2$ $\bullet^2$✘ | $k\cos B = 5,\ k\sin B = 2$ $\bullet^2$ ✓1 |
| $\tan a = \frac{2}{5},\ a = 0\cdot3805...$ | $\tan x = \frac{2}{5},\ x = 0\cdot3805...$ | $\tan B = \frac{2}{5},\ B = 0\cdot3805...$ |
| $\sqrt{29}\cos(x+0\cdot38)$ $\bullet^3$✓ $\bullet^4$✓ | $\sqrt{29}\cos(x+0\cdot38)$ $\bullet^3$✓ $\bullet^4$ ✓1 | $\sqrt{29}\cos(x+0\cdot38)$ $\bullet^3$✓ $\bullet^4$ ✓1 |

| | | | | |
|---|---|---|---|---|
| | (b) | $\bullet^5$ equate to 12 and simplify constant terms<br>$\bullet^6$ use result of part (a) and rearrange<br>$\bullet^7$ solve for $x+a$<br>$\bullet^8$ solve for $x$ | $\bullet^5$ $5\cos x - 2\sin x = 2$ or $5\cos x - 2\sin x - 2 = 0$<br>$\bullet^6$ $\cos(x+0\cdot3805...) = \frac{2}{\sqrt{29}}$<br>$\bullet^7$ $1\cdot1902...$, $\bullet^8$ $5\cdot0928...$<br>$\bullet^7$ $0\cdot8097...$, $\bullet^8$ $4\cdot712...$ | **4** |

**Notes:**

10. The values of $x$ may be given in radians or degrees.
11. Do not penalise candidates who attribute the values of $x$ to the wrong points.
12. Accept any answers, in degrees or radians, that round correct to one decimal place.
13. $\bullet^4$ is unavailable for candidates who give their answer in degrees in part (a) and in part (b). $\bullet^4$ is unavailable for candidates who give their answer in degrees in part (a) and radians in part (b). $\bullet^8$ is unavailable for candidates who give their answer in radians in part (a) and degrees in part (b).

**Conversion Table:**

| Degrees | Radians |
|---|---|
| $21\cdot8$ | $0\cdot3805...$ |
| $46\cdot4$ | $0\cdot8097...$ |
| $68\cdot2$ | $1\cdot190...$ |
| $270$ | $4\cdot712...$ or $\frac{3\pi}{2}$ |
| $291\cdot8$ | $5\cdot0928...$ |

**Commonly Observed Responses:**



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| Question | Generic Scheme | Illustrative Scheme | Max Mark |
|---|---|---|---|
| **9.** | $\bullet^1$ write in integrable form | $\bullet^1$ $2x^{\frac{1}{2}} + x^{-\frac{1}{2}}$ | **4** |
| | $\bullet^2$ integrate one term | $\bullet^2$ $\frac{4}{3}x^{\frac{3}{2}}$ or $2x^{\frac{1}{2}}$ | |
| | $\bullet^3$ complete integration | $\bullet^3$ $2x^{\frac{1}{2}} + c$ or $\frac{4}{3}x^{\frac{3}{2}} + c$ | |
| | $\bullet^4$ state expression for $f(x)$ | $\bullet^4$ $f(x) = \frac{4}{3}x^{\frac{3}{2}} + 2x^{\frac{1}{2}} - 2$ | |

**Notes:**

1. For candidates who do not attempt to write $f'(x)$ as the sum of two integrable terms, award 0/4.
2. $\bullet^2$ and $\bullet^3$ are only available for integrating terms involving fractional indices.
3. The term integrated at $\bullet^3$ must have an index of opposite sign to that of the term integrated at $\bullet^2$.
4. For candidates who differentiate one term, only $\bullet^1$ and $\bullet^2$are available.
5. For candidates who differentiate both terms, only $\bullet^1$ is available.
6. For $\bullet^4$ . accept ' $f(x) = \frac{4}{3}x^{\frac{3}{2}} + 2x^{\frac{1}{2}} + c,\ c = -2$ '
7. Candidates must simplify coefficients in **their** final line of working for the last mark available for that line of working to be awarded.

**Commonly Observed Responses:**

**Candidate A**

$f'(x) = 2x + x^{-\frac{1}{2}}$ $\bullet^1$ ✗

$x^2 + 2x^{\frac{1}{2}} + c$ $\bullet^2$ ✓ $\bullet^3$ ✓2

$f(x) = x^2 + 2x^{\frac{1}{2}} - 47$ $\bullet^4$ ✓1

**Candidate B**

$f'(x) = 2x + x^{-\frac{1}{2}}$ $\bullet^1$ ✗

$x^2 + 2x^{\frac{1}{2}}$ $\bullet^2$ ✓ $\bullet^3$ ✗

$f(x) = x^2 + 2x^{\frac{1}{2}}$ $\bullet^4$ ^

**Candidate C**

$f'(x) = 2x^{\frac{1}{2}} + 1$ $\bullet^1$ ✗

$\frac{4}{3}x^{\frac{3}{2}} + x + c$ $\bullet^2$ ✓ $\bullet^3$ ✓2

$f(x) = \frac{4}{3}x^{\frac{3}{2}} + x - 5$ $\bullet^4$ ✓1

**Candidate D**

$f'(x) = \frac{2x+1}{x^{\frac{1}{2}}}$ $\bullet^1$ ^

$\frac{x^2 + x}{2x^{\frac{3}{2}}} + c$ $\bullet^2$ ✗ $\bullet^3$ ✗ $\bullet^4$ ✗

See Note 1

$f(x) = \frac{x^2 + x}{2x^{\frac{3}{2}}} + \frac{115}{3}$

| Question | Generic Scheme | Illustrative Scheme | Max Mark |
|---|---|---|---|
| **Candidate E**<br>$f'(x) = 2x^{\frac{1}{2}} + x^{-\frac{1}{2}}$ ●1 ✓<br>$= \frac{2x^{\frac{3}{2}}}{\frac{3}{2}} + \frac{x^{\frac{1}{2}}}{\frac{1}{2}} + c$ ●2 ✓<br>●3 ✓2 ●4 ^ | | | |
| **10.** (a) | ●1 Start to differentiate<br>●2 Complete differentiation | ●1 $\frac{1}{2}(x^2+7)^{-\frac{1}{2}}$.....<br>●2 .....$\times 2x$ | **2** |
| **Notes:** | | | |
| 1. On this occasion there is no requirement to simplify coefficients. | | | |
| **Commonly Observed Responses:** | | | |
| | | | |
| (b) | ●3 link to (a) and integrate | ●3 $4(x^2+7)^{\frac{1}{2}}\ (+c)$ | **1** |
| **Notes:** | | | |
| 2. A candidate's answer at ●3 must be consistent with earlier working. | | | |
| **Commonly Observed Responses:** | | | |
| **Candidate A**<br>$\int 4x(x^2+7)^{\frac{-1}{2}}dx$<br>$= \frac{4x(x^2+7)^{\frac{1}{2}}}{\frac{1}{2}\times 2x} + c$<br>$= \frac{4x(x^2+7)^{\frac{1}{2}}}{x} + c$<br>$= 4(x^2+7)^{\frac{1}{2}} + c$ ●3 × | | | |

| Question | Generic Scheme | Illustrative Scheme | Max Mark |
|---|---|---|---|
| **11.** (a) | $\bullet^1$ substitute for $\sin 2x$ and $\tan x$ | $\bullet^1$ $(2\sin x\cos x)\times\dfrac{\sin x}{\cos x}$ | **4** |
| | $\bullet^2$ simplify | $\bullet^2$ $2\sin^2 x$ | |
| | $\bullet^3$ use an appropriate substitution | $\bullet^3$ $2\left(1-\cos^2 x\right)$ or $1-\left(1-2\sin^2 x\right)$ | |
| | $\bullet^4$ simplify and communicate result | $\bullet^4$ $1-\cos 2x=1-\cos 2x$ or $2\sin^2 x=2\sin^2 x$ $\therefore$ Identity shown | |

**Notes:**

1. $\bullet^1$ is not available to candidates who simply quote $\sin 2x=2\sin x\cos x$ and $\tan x=\dfrac{\sin x}{\cos x}$ without substituting into the identity.
2. $\bullet^4$ is not available to candidates who work throughout with A in place of $x$.
3. $\bullet^3$ is not available to candidates who simply quote $\cos 2x=1-2\sin^2 x$ without substituting into the identity.
4. On this occasion, at $\bullet^4$ do not penalise the omission of 'LHS = RHS' or a similar statement.

**Commonly Observed Responses:**

**Candidate A**

$\sin 2x\tan x=1-\cos 2x$

$2\sin x\cos x\times\dfrac{\sin x}{\cos x}=1-\cos 2x$ $\bullet^1$✓

$2\sin^2 x=1-\cos 2x$ $\bullet^2$✓

$2\sin^2 x-1=-\cos 2x$ $\bullet^3$✗ $\bullet^4$✗

$-(1-2\sin^2 x)=-\cos 2x$

$-\cos 2x=-\cos 2x$

In proving the identity, candidates must work with both sides independently. ie in each line of working the LHS must be equivalent to the left hand side of the line above.

**Candidate B**

$\sin 2x\tan x=1-\cos 2x$

$\sin 2x\tan x=1-(1-2\sin^2 x)$ $\bullet^1$^ $\bullet^2$^

$\sin 2x\tan x=2\sin^2 x$ $\bullet^3$✓

$\tan x=\dfrac{2\sin^2 x}{2\sin x\cos x}$ $\bullet^4$✗

$\tan x=\tan x$

| Question | Generic Scheme | Illustrative Scheme | Max Mark |
|---|---|---|---|
| (b) | •5 link to (a) and substitute<br><br>•6 differentiate | •5 $f(x) = 1 - \cos 2x$ or $f(x) = 2\sin^2 x$<br><br>•6 $f'(x) = 2\sin 2x$ or $f'(x) = 4\sin x \cos x$ | 2 |
| **Notes:** | | | |
| | | | |
| **Commonly Observed Responses:** | | | |
| | | | |

**[END OF MARKING INSTRUCTIONS]**